

Harini Badam Data Engineer

Python | SQL | AWS | GCP | Apache Spark | dbt | Airflow | Snowflake

Harini@mycareermail.com | (601) 913-3887 | USA | LinkedIn

Summary

Detail-oriented and results-driven Data Engineer with around 4 years of experience in building and optimizing scalable data pipelines, cloud-native data platforms, and real-time analytics solutions. Skilled in developing robust ETL/ELT workflows using Python, SQL, dbt, and Airbyte, and proficient in leveraging cloud ecosystems (GCP, AWS) and big data frameworks (Kafka, Spark, Hadoop). Experienced in data modeling, orchestration, and machine learning, with added exposure to tools like Snowflake, Looker, PostgreSQL, BigQuery, and AI-powered solutions to enable intelligent business decisions.

Technical Skills

Programming Languages: Python, SQL, Java

Cloud Platforms: AWS (S3, Glue, Redshift, Lambda), GCP (BigQuery, Dataproc, Cloud Storage)

Data Engineering & Big Data Frameworks: Apache Spark, Apache Kafka, Apache Hive, Hadoop (HDFS), Databricks, Apache Airflow, Airbyte

ETL & Orchestration Tools: dbt, AWS Glue, Airflow, Terraform, GitHub Actions, Data Lake Architecture

Databases & Data Warehousing: Snowflake, PostgreSQL, BigQuery, Redshift, DynamoDB

Business Intelligence & Visualization: Power BI, Tableau, Looker

Machine Learning & AI: scikit-learn, TensorFlow, Hugging Face, OpenAI API, NLP & Computer Vision Basics

Infrastructure & DevOps: Docker, Kubernetes (EKS), Zookeeper, Infrastructure as Code (Terraform)

Data Governance & Metadata Management: HBase, Data Quality Auditing, Metadata Cataloging

Professional Experiences

Data Engineer, Wells Fargo

01/2024 – Present | USA

- Architected a scalable data lake on GCP using Cloud Storage and BigQuery, managing over 5 TB of multi-format structured and unstructured data. Designed data zones (raw, staging, curated) to support ML-ready analytics pipelines.
- Developed real-time streaming data pipelines using Apache Kafka with Zookeeper, handling 10K+ events per second and enabling fault-tolerant data ingestion across multiple banking systems.
- Built and maintained distributed batch jobs in PySpark on Databricks, processing over 2 million records per day. Used Pandas and advanced SQL (JOINS, CTEs, Window Functions) for data transformation and enrichment.
- Developed ML models in TensorFlow and scikit-learn for customer churn and fraud detection. Integrated model inference in pipelines using Airflow, improving model deployment timelines and increasing accuracy by 25%.
- Improved reporting by optimizing Hive queries on Dataproc and offloading complex aggregations to BigQuery, leading to a 40% reduction in report generation time.
- Modeled curated datasets with dbt, enforcing modular, testable, and version-controlled data transformations; integrated Airbyte for ingestion from API and SaaS connectors.
- Spearheaded implementation of metadata governance using HBase, enabling automated lineage tracking and cataloging for 50+ analysts and data scientists.
- Partnered with BI teams to build Power BI and Looker dashboards, driving a 30% improvement in adoption of analytics tools for executive and operational decision-making.

Data Engineer, Tyler Technologies

05/2023 – 12/2023 | USA

- Engineered event-driven streaming pipelines with Apache Kafka and Spark Streaming to handle inter-departmental communication across governmental platforms, increasing data availability speed by 41%.

- Designed serverless ETL workflows in AWS Glue for data extraction and transformation across S3, Redshift, and DynamoDB; incorporated dbt for transformation and documentation layers.
- Built ingestion connectors using Airbyte to extract data from REST APIs and third-party services. Scheduled workflows with Airflow to enable robust monitoring and failure recovery.
- Integrated Snowflake as a centralized data warehouse, implementing clustering keys and materialized views for performance optimization. Connected PostgreSQL for transactional dataset offloading.
- Led deployment automation using Terraform and GitHub Actions for CI/CD, achieving 35% faster release cycles and reducing human errors in infrastructure provisioning.
- Deployed Docker containers orchestrated with Kubernetes (EKS) for running high-performance data processing tasks, improving platform stability and resource efficiency.
- Built internal APIs in Java to serve pre-aggregated data from Snowflake to downstream systems with authentication and rate-limiting.
- Partnered with analytics teams to roll out real-time dashboards in Looker, enhancing data transparency across state departments.

Data Engineer, Genpact

05/2021 – 12/2022 | India

- Designed and implemented a centralized Hadoop-based data lake using HDFS and Apache Hive to integrate data from 15+ sources, including CRM, ERP, and point-of-sale systems.
- Built modular ETL pipelines using Apache Airflow, Python, and SQL for real-time data syncing, cleansing, and reconciliation across departments.
- Leveraged Apache Spark and dbt to cleanse, model, and transform over 500K daily records, enabling downstream analytical consumption and improving reporting accuracy.
- Created data pipelines for cloud storage using AWS S3 and automated transformation via Glue jobs, reducing storage and compute costs by 25% without compromising speed.
- Built PostgreSQL and Redshift views to aggregate and normalize transactional data, making it readily available for executive dashboards and audit teams.
- Designed interactive Tableau dashboards and SQL-based reporting layers to monitor customer behavior, sales patterns, and warehouse KPIs.
- Ensured compliance with internal governance policies by implementing data quality auditing workflows, custom validations, and daily pipeline health checks.
- Participated in early adoption of AI forecasting models for inventory optimization using OpenAI embeddings and regression models, resulting in improved stock planning cycles.

Education	
Master of Science , <i>Indiana Wesleyan University</i> Information Technology	01/2023 - 08/2024 Florence, KY, USA
Bachelor of Engineering , <i>Narasaraopeta Engineering College</i> Electronics and Communication	07/2018 - 06/2022 Narasaraopet, A.P, India

Certificate	
Great Learning AWS Project – (Link)	
• Gained hands-on experience in deploying cloud solutions on AWS, including configuring services like EC2, S3, and Lambda for real-world business scenarios.	
Great Learning Data Visualization using Power BI – (Link)	
• Learned to create interactive dashboards and visual reports using Power BI, leveraging DAX, data modeling, and custom visuals for insightful decision-making.	